IT IS EXTREMELY IMPORTANT THAT YOU PAY YOUR FULL ATTENTION TO THE EXPERIMENT.

Therefore, we kindly ask you to:

1. **Find a quiet, calm space to do this study.** Please turn off your phone and other media to reduce distractions.

2. **Only participate from your desktop computer.**

3. **Do not exit the experiment once it has begun**. Due to the software we are using, we cannot guarantee you that you will be able to come back to the experiment and that your data can be recorded properly. In that case, we cannot compensate you for your time. Keep in mind the whole study should not take you more than 5 minutes.

4. **Focus on the tasks and respond attentively, but promptly.**

Not following these instructions can affect your compensation. If you fail to respond to 10% or more of the trials in this experiment, you will not receive (full) compensation. Thank you for your time and understanding.

Description: In this survey you will take part in a couple of tasks dealing with your opinions about a robotic dog.

Risks and Benefits: You will not be exposed to any sensitive topics. There appear to be no risks and discomforts associated with the session other than the ones common in everyday life.

Time Involvement: Your participation in this part of the session will take approximately 15 minutes.

Confidentiality and anonymity: All published and written data that may result from these studies will only be shared in an aggregate form and will be anonymous.

Compensation: You will receive credits for this session (exact amount announced in Sona) if you follow up on all instructions. **If you fail to respond to 10% of the trials in this experiment, you will receive 1 credit less.** If you do not finish the survey, you will not receive your credits.

Participation: Participation in this research study is completely voluntary. You have the right to withdraw at anytime or refuse to participate entirely. If you wish to stop participating, no credits will be allocated. You can participate in other sessions instead.

Contact Information: If you have any questions, concerns, or complaints, please contact Femke van Horen (femke.van.horen@vu.nl).

For the following task, we are interested in how you perceive different products.

On each screen, you will see two products.

To get familiar with the task, you will first do a **practice round**.

You will **see two citrus fruits**. We will ask you to indicate which of the two citrus fruits **you like more.**

If you like the fruit on the **left** more, press **D**
If you like the fruit on the **right** more, press **K**

Please focus on the task and answer as fast as you can without thinking too much. We are interested in your initial responses, **there are no right or wrong answers.**

You can position your index fingers on the required keys: **left index finger on D, and the right index finger on K**.

Please note that WE WILL BE CHECKING YOUR ATTENTION throughout the experiment.

In attention check trials, you will see TWO IDENTICAL IMAGES on the screen.

You will be asked to PRESS Q instead of K or D like in other trials.

Attention check trials are NOT TIMED, so you will have enough time to respond.

Now that you are familiar with the task, we continue with the main task.

The main task is about **ROBOTIC ARMS**

A robotic arm is a mechanical device consisting of linked segments and joints. It is designed to mimic the movements and functionalities of a human arm in various medical tasks.

In **rehabilitation**, robotic arms help patients regain movement after injuries such as stroke or broken joints. They provide **controlled, repeatable assistance**, adapting to a patient's progress and enabling high-intensity training.

Please imagine that you have broken your hand and you need rehabilitation to restore mobility and strength. A robotic arm, instead of a physiotherapist, is used to do the exercises with you.

Now take a minute to picture that you will have to hold the robotic hand in order to do the exercises.

In the next three tasks, you will evaluate a set of robotic arms on different dimensions. On each screen, you will see two robotic arms holding the same human hand, and you will be asked to make a judgment.

We now start with the first task. Please read the instructions carefully.

As in the practice trials, you will see two robotic arms holding the same human hand next to each other. Please focus on the **robotic arm**. In this first task, you will be asked to indicate which of the two robotic arms **you like more.**

If you **like** the robotic arm on the **left** more, press **D**
If you **like** the robotic arm on the **right** more, press **K**

Please focus on the task and answer as fast as you can without thinking too much. We are interested in your initial responses; there are no right or wrong answers.

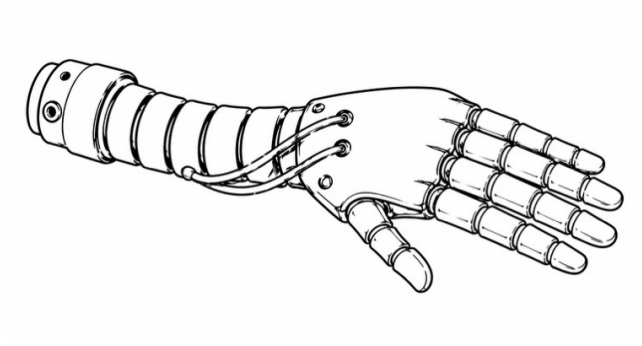
You can position your index fingers on the required keys: **left index finger on D, and the right index finger on K**.

Please note that WE WILL BE CHECKING YOUR ATTENTION throughout the experiment!

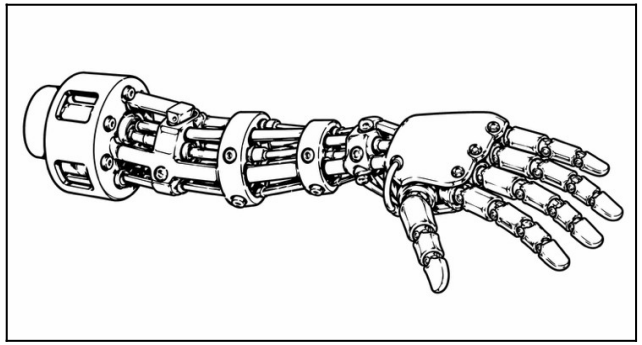
Before the task begins, you will first see what the task looks like.

Your task is to decide which of the two robotic arms **you LIKE MORE.**

ROBOTIC ARM A



ROBOTIC ARM B



If you like the left one more
(ROBOTIC ARM A),
you **PRESS D**

If you like the right one more
(ROBOTIC ARM B),
you **PRESS K**

For Jeroen: On the next pages the randomized trials are shown. Please keep the following text each time above the display of the two arms:

Imagine you need to do rehabilitation exercises with a robotic hand. Your task is to decide which of the two robotic arms you **LIKE MORE** to hold**.**

Thank you, the first task is finished!

Now, we will again show you the same robotic arms. This time it is your task to indicate which of the two robotic arms you think is more **EERIE**.

Eerie refers to a feeling of unease or discomfort that is unsettling (strange, unnatural).

If the robotic arm on the **left** feels more **eerie**, press **D**
If the robotic arm on the **right** feels more **eerie**, press **K**

You can position your index fingers on the required keys: **left index finger on D, and the right index finger on K**.

For Jeroen: On the next pages the randomized trials are shown. Please keep the following text each time above the display of the two arms:

Imagine you need to do rehabilitation exercises with this robotic hand. Your task is to decide which of the two robotic arms **feels more EERIE.**

Thank you, the second task is finished!

Now, we will again show you the same robotic arms. This time it is your task to indicate which of the two robotic arms is **more similar to a human arm**.

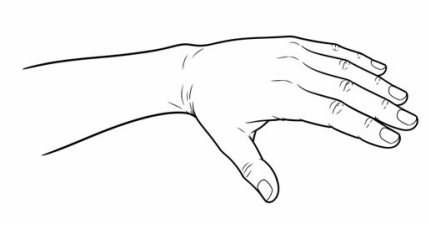
If the robotic arm on the **left** is more similar, press **D**
If the robotic arm on the **right** is more similar, press **K**

Please focus on the task and answer as fast as you can without thinking too much. We are interested in your initial responses, there are no right or wrong answers.

Before the task begins, you will first see what the task looks like.

Your task is to decide which of the two robotic arms at the bottom is

MORE SIMILAR TO THE HUMAN ARM ON TOP.

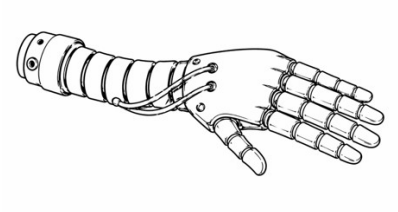


If you think the left one (ROBOTIC ARM A), is more similar

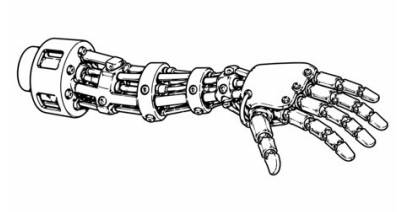
you **PRESS D.**

If you think the right one (ROBOTIC ARM B), is more similar

you **PRESS K.**



ROBOTIC ARM A



ROBOTIC ARM B

For Jeroen: On the next pages the randomized trials are shown. Please keep the following text each time above the display of the two arms:

Your task is to decide which of the two robotic arms **is more SIMILAR to the human arm on top.**

Demographics

- Age
- Gender
- To what extent are you familiar with robotic arms in health care (1 not at all – 7 very much)

  ⇨ Jeroen: underneath you find seveal personality characteristics scales. I am not sure if we should program them in the psychopy experiment or in a separate qualtrics experiment. If the first, it is important that the different items are indicated clearly (maybe alternating grey vs white colored items?), so students do not get bored, and can read the different items easily. If we choose for the last option, we will need to make sure that the data can be linked. So the students need to indicate the correct ID number twice. I am not sure if that is too risky. Can you redirect the experimetn to a qualtrics experiment or is that difficult?

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Provide instructions:
Underneath you will find several statements. Please indicate the extent in which you agree with the statements, from 1 = do not agree at all to 7 = totally agree. There are no right or wrong answers, we are interested in your opinion.

1-7 likert scale totally disagree to totally agree.

2. It is absurd to consider a human being and a robot to be the same kind of thing. (.78)
3. Human beings have a soul, which a robot could never have. (.77)
4. Even if a robot might one day seem human, it would never be anything like a real human being. (.77)
6. In a sense, human beings are nothing more than highly sophisticated, self-replicating robots.R (.73)
7. It would be alright if someday we could not tell robots from human beings.R (.72)
8. Human beings are fundamentally different from robots. (.72)

Next page

Technology anxiety (general anxiety with all forms of technology and were assessed on a seven-point likert scale with endpoints strongly agree to strongly disagree. Higher scores indicate higher levels of anxiety). Meuter, M. L., Ostrom, A. L., Bitner, M. J., & Roundtree, R. (2003). The influence of technology anxiety on consumer use and experiences with self-service technologies. *Journal of business research*, *56*(11), 899-906. => items 1-3.

1. I feel apprehensive about using technology.
2. I have avoided technology because it is unfamiliar to me.
3. I am able to keep up with important technological advances.
4. I feel anxious when using new technologies.

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Full disgust sensitivity scale

Appendix

Disgust Scale—Revised (DS–R): Items, Scaling, and Scoring

DS–R Part I: Please circle true or false	Scaling		
1. I might be willing to try eating monkey meat, under some circumstances. (R)	False		True
2. It would bother me to see a rat run across my path in a park.	False		True
3. Seeing a cockroach in someone else's house doesn't bother me. (R)	False		True
4. It bothers me to hear someone clear a throat full of mucus.	False		True
5. If I see someone vomit, it makes me sick to my stomach.	False		True
6. It would bother me to be in a science class, and see a human hand preserved in a jar.	False		True
7. It would not upset me at all to watch a person with a glass eye take the eye out of the socket. (R)	False		True
8. It would bother me tremendously to touch a dead body.	False		True
9. I would go out of my way to avoid walking through a graveyard.	False		True
10. I never let any part of my body touch the toilet seat in a public washroom.	False		True
11. I probably would not go to my favorite restaurant if I found out that the cook had a cold.	False		True
12. Even if I was hungry, I would not drink a bowl of my favorite soup it if had been stirred with a used but thoroughly washed flyswatter.	False		True
13. It would bother me to sleep in a nice hotel room if I knew that a man had died of a heart attack in that room the night before.	False		True
DS–R Part II: Please rate how disgusting you would find the following experiences.			
14. If you see someone put ketchup on vanilla ice cream and eat it.	Not	Slightly	Very
15. You are about to drink a glass of milk when you smell that it is spoiled.	Not	Slightly	Very
16. You see maggots on a piece of meat in an outdoor garbage pail.	Not	Slightly	Very
17. You are walking barefoot on concrete and step on an earthworm.	Not	Slightly	Very
18. While you are walking through a tunnel under a railroad track, you smell urine.	Not	Slightly	Very
19. You see a man with his intestines exposed after an accident.	Not	Slightly	Very
20. Your friend's pet cat dies and you have to pick up the dead body with your bare hands.	Not	Slightly	Very
21. You accidentally touch the ashes of a person who has been cremated.	Not	Slightly	Very
22. You take a sip of soda and realize that you drank from the glass that an acquaintance of yours had been drinking from.	Not	Slightly	Very
23. You discover that a friend of yours changes underwear only once a week.	Not	Slightly	Very
24. A friend offers you a piece of chocolate shaped like dog-doo.	Not	Slightly	Very
25. As part of a sex education class, you are required to inflate a new lubricated condom, using your mouth.	Not	Slightly	Very

Note. *Scoring:* Items with (R) are reverse scored; *True* = 1, *False* = 0; *Not disgusting* = 0, *Slightly disgusting* = 0.5, *Very disgusting* = 1. A total disgust sensitivity score can be calculated by summing responses to the 25 items (0–25). Scores for each of three subscales are calculated as follows. Core Disgust: sum of items 1, 2, 3, 4, 5, 12, 14, 15, 16, 17, 18, and 23; Animal Reminder Disgust: sum of items 6, 7, 8, 9, 13, 19, 20, and 21; Contamination-Based Disgust: sum of items 10, 11, 22, 24, and 25. Items are from "Individual Differences in Sensitivity to Disgust: A Scale Sampling Seven Domains of Disgust Elicitors," by J. Haidt, C. McCauley, and P. Rozin, 1994, *Personality and Individual Differences, 16,* 700–713. Copyright 1994 by Elsevier.

TABLE 2
The Revised 12-Item Religious Fundamentalism Scale

This survey is part of an investigation of general public opinion concerning a variety of social issues. You will probably find that you *agree* with some of the statements, and *disagree* with others, to varying extents. Please indicate your reaction to each statement by blackening a bubble in SECTION 1 of the bubble sheet, according to the following scale:

Blacken the bubble labeled –4 if you *very strongly disagree* with the statement.
–3 if you *strongly disagree* with the statement.
–2 if you *moderately disagree* with the statement
–1 if you *slightly disagree* with the statement.
Blacken the bubble labeled +1 if you *slightly agree* with the statement.
+2 if you *moderately agree* with the statement.
+3 if you *strongly agree* with the statement.
+4 if you *very strongly agree* with the statement.
If you feel exactly and precisely *neutral* about an item, blacken the "0" bubble.

You may find that you sometimes have different reactions to different parts of a statement. For example, you might very strongly disagree ("–4") with one idea in a statement, but slightly agree ("+1") with another idea in the same item. When this happens, please combine your reactions, and write down how you feel on balance (a "–3" in this case).

1. God has given humanity a complete, unfailing guide to happiness and salvation, which must be totally followed.
2. No single book of religious teachings contains all the intrinsic, fundamental truths about life.[a]
3. The basic cause of evil in this world is Satan, who is still constantly and ferociously fighting against God.
4. It is more important to be a good person than to believe in God and the right religion.[a]
5. There is a particular set of religious teachings in this world that are so true, you can't go any "deeper" because they are the basic, bedrock message that God has given humanity.
6. When you get right down to it, there are basically only two kinds of people in the world: the Righteous, who will be rewarded by God; and the rest, who will not.

any "deeper" because they are the basic, bedrock message that God has given humanity.

6. When you get right down to it, there are basically only two kinds of people in the world: the Righteous, who will be rewarded by God; and the rest, who will not.
7. Scriptures may contain general truths, but they should NOT be considered completely, literally true from beginning to end.[a]
8. To lead the best, most meaningful life, one must belong to the one, fundamentally true religion.
9. "Satan" is just the name people give to their own bad impulses. There really is *no such thing* as a diabolical "Prince of Darkness" who tempts us.[a]
10. Whenever science and sacred scripture conflict, *science* is probably right.[a]
11. The fundamentals of God's religion should never be tampered with, or compromised with others' beliefs.
12. *All* of the religions in the world have flaws and wrong teachings. There is *no* perfectly true, right religion.[a]

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[a] indicates item is worded in the con-trait direction, for which the scoring key is reversed.